

බස්නාහිර පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව Department of Education – Western Province			
පළමු වාර ඇගයීම } 2019 First Term Evaluation			
ශ්‍රේණිය } 10 Grade	විෂයය } Science Subject	පත්‍රය } I Paper	කාලය } 01 hour Time
නම } Name		විභාග අංකය } Index Number	

Answer all the questions. Underline the correct or most suitable answer.

1. Which of the following is a vector quantity?
 (1) Distance (2) Speed (3) Time (4) Velocity
2. Beri – beri is caused due to the deficiency of vitamin which is essential for the formation of bone marrow and lipid metabolism. What is this vitamin?
 (1) Vitamin A (2) Vitamin D (3) Vitamin C (4) Vitamin B
3. Which of the following is known as tonoplast?
 (1) Vacuole membrane (2) Cell wall (3) Plasma membrane (4) Nuclear membrane
4. The number of protons, neutrons and electrons in $^{35}_{17}\text{Cl}^-$ respectively?
 (1) 17,18, 17 (2) 17,18,18 (3) 17,17,18 (4) 17,18,16
5. What is the main function of ribosomes?
 (1) production of secretory vesicles (2) Protein synthesis
 (3) Transportation of proteins (4) production of energy
6. Which of the following animal has 46 chromosomes inside the nucleus?
 (1) man (2) frog (3) sparrow (4) fly
7. The compound of which metal is used to produce antacid?
 (1) Mg (2) Na (3) Ca (4) K

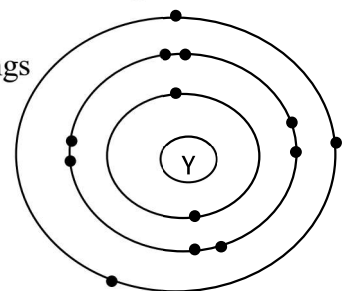
Answer the question Number 8 and 9 in relation to the atomic structure of the element Y given below.

8. Select the correct answer that shows the period and group to which Y belongs

- (1) period 2 , Group III (2) Period 3, Group III
 (3) Period 2, Group II (4) Period 4 , Group III

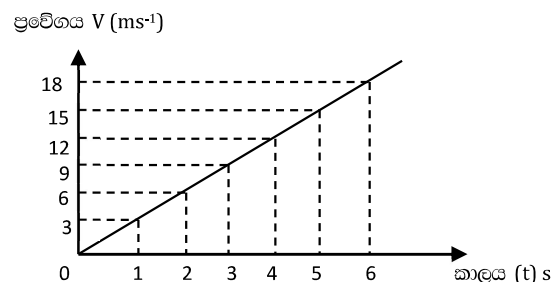
9. What could be the element denoted by the letter Y

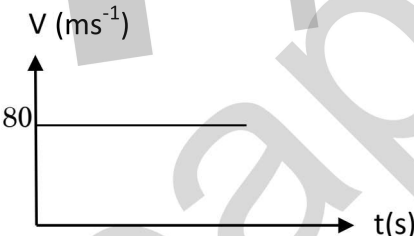
- (1) Mg (2) K (3) Ca (4) Al



10. Velocity time graph of a certain motion is shown in the figure below. The displacement in 6s is

- (1) 27m
 (2) 45m
 (3) 54m
 (4) 108m



11. (A) Has a magnitude. (B) Has a definite direction.
(C) No definite magnitude. (D) No definite direction.
The correct statement in relation to velocity from A,B,C and D above
(1) Only A (2) only A and B (3) Only C and D (4) Only A and C
12. Which of the following ion has the same number of electron as Na^+
(1) Cl^- (2) O^{2-} (3) Mg^+ (4) Ca^{2+}
13. Electronic configuration of some elements is given below. Given symbols are not standard symbols.
A – 2,1 B – 2,3 C – 2,6 D – 2,7
Which of the following answer gives the valency of each element correctly?
(1) 1,3,2,1 (2) 1,3,6,7 (3) 1,3,1,2 (4) 1,5,6,7
14. Consider the situations given below.
A- Grooves are etched on the surface of tires.
B - Applying lubricators between the contact surfaces.
C – Inserting ball bearing and roller bearing between the contact surfaces.
What are the instance / s of increasing the friction between the contact surfaces from the above?
(1) Only A (2) only B (3) Only A and B (4) Only B and C
15. Three characteristics of a certain element are given below.
• Does not react with cold water, but reacts with hot water.
• Burns in air with bright white flame.
• Reacts with dilute acids and liberate hydrogen gas.
(1) Na (2) C (3) Mg (4) S
16. 
The graph shows ,
(1) Motion of an object moving at uniform acceleration.
(2) Motion of an object moving at uniform deceleration.
(3) Motion of an object moving at uniform velocity.
(4) velocity time graph of an object at rest.
17. Which of the following organelle helps in the maintenance of water balance and support in the cell?
(1) Cell wall (2) cytoplasm (3) Golgi bodies (4) Vacuole
18. The process by which new cells are formed by the division is known as
(1) Cell differentiation (2) specialization (3) cell division (4) Regeneration
19. What is the momentum of an object of mass 2000000g moving at a velocity 20ms^{-1} ?
(1) 40 kgms^{-1} (2) 4000 kgms^{-1} (3) 40000kgms^{-1} (4) 40000000kgms^{-1}

20. Which of the following is the correct formulae of the compound that contains only P and H?

- (1) PH (2) PH₂ (3) PH₃ (4) PH₄

21. The correct chemical formulae of Calcium phosphate is

- (1) CaPO₄ (2) Ca₃PO₄ (3) Ca₂(PO₄)₃ (4) Ca₃(PO₄)₂

22.



The diagrams above show organelles present in cells. What is the name of the organelles A and B

- (1) Mitochondrion, Golgi complex (2) Golgi complex, Rough endoplasmic reticulum
(3) Nucleus, Mitochondrion (4) Endoplasmic reticulum, Golgi complex

23. Which of the following gives the correct expression of momentum?

- (1) mv (2) m/v (3) v/m (4) m²v

24. Which of the following statement/s describe the Newton's second law?

- a. Acceleration of an object is directly proportional to the unbalanced force.
b. Acceleration of an object is inversely proportional to the unbalanced force.
c. Acceleration is directly proportional to the mass.
d. Acceleration is inversely proportional to the mass.

- (1) only a and d (2) only a and c (3) only b and c (4) only a, b, c and d

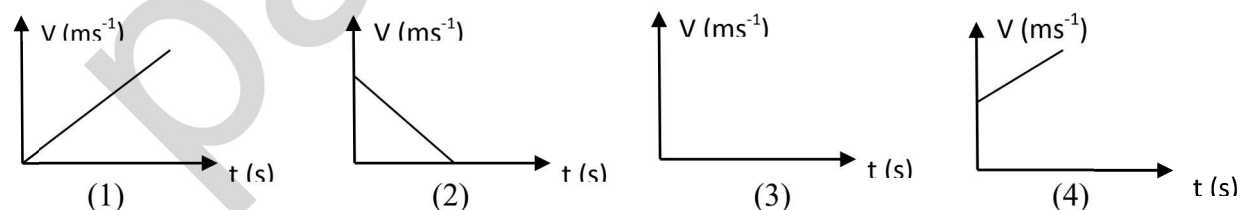
25. Chemical formulae of a compound is X₂Y. Which of the following pair of elements are suitable for X and Y respectively?

- (1) Mg, Cl (2) K, O (3) Mg, O (4) K, Cl

26. What is the standard method of representing Deuterium Isotope?

- (1) ${}^1_1\text{H}$ (2) ${}^2_1\text{H}$ (3) ${}^3_1\text{H}$ (4) ${}^2_2\text{H}$

27. What is the velocity time graph for a fruit falling from a tree?



28. What is the element which is a constituent of amino acids and proteins that causes chlorosis in veins and between areas of veins?

- (1) Sulphur (2) Iron (3) Calcium (4) Zinc

29. Which of the following elements would be the examples for metals, metalloids and non-metals respectively

- (1) Na, Al, B (2) Mg, Ne, C (3) Na, Si, C (4) N, O, F

30. If an object gains an acceleration of 4ms⁻² due a force of 16N, calculate the mass of the object.

- (1) 0.25 kg (2) 4 kg (3) 16 kg (4) 64 kg

31. If the velocity of an object decreases uniformly from 16ms^{-1} to 4ms^{-1} , calculate the deceleration of the object.

- (1) -3ms^{-2} (2) -3ms^{-1} (3) 3ms^{-2} (4) 3ms^{-1}

32. An object has been projected vertically upwards with an initial velocity of 30ms^{-1} . Calculate the maximum height reached by the ball.

- (1) 10 m (2) 30m (3) 35m (4) 45m

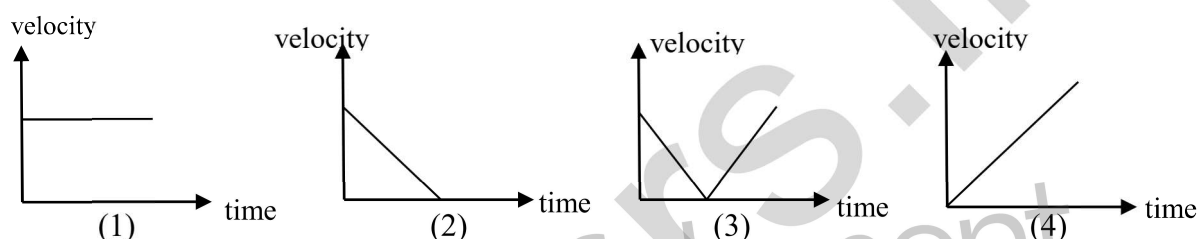
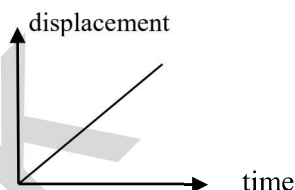
33. The electronic configuration of a certain element 'X' is 2,8,8,1. What would be this element?

- (1) Ca (2) Mg (3) K (4) Na

34. Which of the following factor affects the limiting frictional force from the factors given below?

- (1) Nature of the contact surface (2) mass of the contact object
(3) Volume of the contact object (4) Area of the contact surface

35. The displacement time graph for the motion of an object is given by the graph. Which of the following graph correctly represent the relevant velocity time graph of it?



36. An object of 5kg falls vertically to the ground. What is the vector quantity that remains constant in this situation?

- (1) Velocity (2) Displacement (3) momentum (4) acceleration

37. Consider the following statement.

- A. The number of protons in isotopes is equal.
B. The number of electrons in isotopes is not equal.
C. The number of neutrons in isotopes are is not equal.

What is the correct answer from the above statement/s

- (1) only A (2) Only B (3) Only C (4) Only A and C

38. Irreversible increase of the dry weight or size of the cell is known as

- (1) Cell development (2) Cell growth (3) Cell differentiation (4) cell specialisations

39. In which form does carbohydrate get stored in animal body?

- (1) As starch (2) As cellulose (3) As glycogen (4) As lactose

40. If the velocity of an object increases from 10ms^{-1} to 25ms^{-1} in 5 seconds, calculate the acceleration of the object.

- (1) 3ms^{-2} (2) 4ms^{-2} (3) 5ms^{-2} (4) -3ms^{-2}